

Bounds of Validity for Bifurcations of Equilibria in a Class of Networked Dynamical Systems

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Introduction to Bifurcations

Definition

Topological changes in the flow (equilibria, stability, oscillations, chaos) when system parameter(s) cross a threshold are called **bifurcations**.

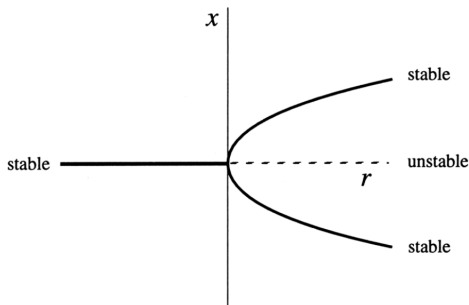


Figure: pitchfork bifurcation diagram for $\dot{x} = rx - x^3$.

Bifurcations in Networked Systems

Many real-world networked systems exhibit *sudden qualitative shifts* in behaviour, which can be modelled as bifurcations:

- **Opinion dynamics:** a population suddenly reaching consensus
- **Network epidemics:** transition from disease-free to endemic state
- **Gene regulatory networks:** switching between cell states

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The **Lyapunov–Schmidt reduction** used to predict and engineer these shifts, but it is inherently local with the validity region typically unknown.

How far from the bifurcation point can we trust the reduction?

Hopfield and Firing Rate Models

System Definition

We study nonlinear network systems of the form:

$$\textbf{Hopfield} : \dot{x} = \Psi(x, p) = -Cx + pAS(x) + b \quad (1a)$$

$$\textbf{Firing Rate} : \dot{x} = \Psi(x, p) = -Cx + S(pAx + b) \quad (1b)$$

with constant **bias** b , **adjacency** matrix A , diagonal **dissipation** matrix $C \succ 0$, nonlinear element-wise **activation** $S(z) = [S_1(z_1), \dots, S_n(z_n)]^\top$. Standard assumptions made about the existence of a bifurcation point, and non-degeneracy in eigenvalues of jacobian $J = D_x \Psi(x^*, p^*)$.

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Applications: modelling of *dynamic neural networks*, *opinion dynamics*, *Lur'e feedback systems*, *gene regulatory networks*, etc.

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Objective: derive **explicit bounds** for the validity region of the LSR for these two classes of models.

Lyapunov–Schmidt reduction

- Bifurcations in high-dimensional systems typically occur on low-dimensional **centre manifold**.
- **L.S. Reduction**: projecting dynamics onto lower-dimensional critical subspace at bifurcation points; **preserves local topological equivalence**.
- **Advantage**: *no explicit centre manifold computation*; direct use of singularity-theoretic tools [GS85].

Lyapunov–Schmidt reduction: introduction

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Output: reduced algebraic system with bifurcation diagram **locally isomorphic** to original system (*equilibria in bijective correspondence*).

Lyapunov–Schmidt reduction: overview of algorithm

- Smooth NLTI system $\dot{\mathbf{x}} = \Phi(\mathbf{x}, \lambda)$
- Singular point at $(\mathbf{x}^*, \lambda^*)$:
 - $\Phi(\mathbf{x}^*, \lambda^*) = 0$
 - For $J = D_{\mathbf{x}}\Phi(\mathbf{x}^*, \lambda^*)$,
 $q := \dim(\ker(J)) > 0$
- Linear orthogonal projection operator $P : \mathbb{R}^n \rightarrow \text{range}(J)$

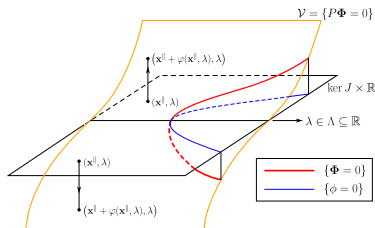
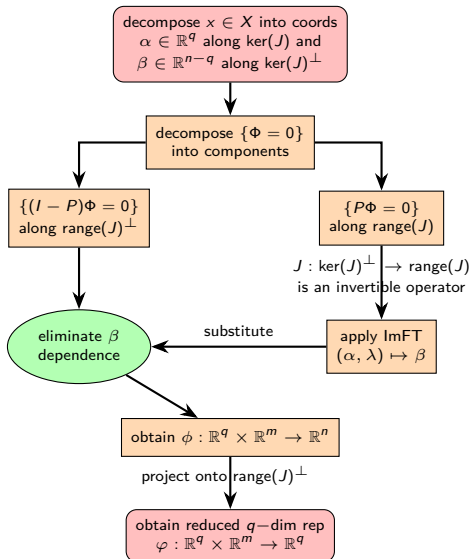


Figure: sketch of LSR for a 2 – dim system with a pitchfork bifurcation



Implicit Function theorem and Bounds

Implicit Function theorem

System $f(u, v) = f(u_0, v_0)$ with invertible linearisation $D_v f(u_0, v_0)$ has a *local unique smooth implicit representation* $v = g(u)$ near (u_0, v_0) .

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Theorem on Bounds for ImFT

Restating from [JCB24, Corollary 3.8], define the following quantities:

- $M_u := \|D_u f(u_0, v_0)\|$ and $M_v := \|(D_v f(u_0, v_0))^{-1}\|$
- $L_u(r_u) := \sup_{u \in \mathcal{B}(u_0, r_u)} \|D_u f(u, v_0) - D_u f(u_0, v_0)\|$
- $L_v(r_u, r_v) := \sup_{(u, v) \in \mathcal{B}(u_0, r_u) \times \mathcal{B}(v_0, r_v)} \|D_v f(u, v) - D_u f(u_0, v_0)\|$

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Then, for all $r_u, r_v > 0$ simultaneously satisfying

$$L_u(r_u)r_u + L_v(r_u, r_v)r_v < \frac{r_v}{M_v} - M_u r_u$$

The implicit map g is $\mathcal{C}^1$ and valid over $g : \mathcal{B}(u_0, r_u) \rightarrow \mathcal{B}(v_0, r_v)$.

Lyapunov–Schmidt reduction: explicit bounds

Theorem on Bounds for LSR [GBB24]

Let $M_u, M_v, r_u, r_v, L_u, L_v \rightarrow M_{\parallel}, M_{\perp}, r_{\parallel}, r_{\perp}, L_{\parallel}, L_{\perp}$ denote the recasts of corresponding quantities into the Lyapunov–Schmidt reduction framework.

Then, for all $r_{\parallel}, r_{\perp} > 0$ simultaneously satisfying

$$L_{\parallel}(r_{\parallel})r_{\parallel} + L_{\perp}(r_{\parallel}, r_{\perp})r_{\perp} < \frac{r_{\perp}}{M_{\perp}} - M_{\parallel}r_{\parallel}$$

the implicit map $(\alpha, \lambda) \xrightarrow{\varphi} \beta$ is valid over $\varphi : \mathcal{B}((\alpha_0, \lambda_0), r_{\parallel}) \rightarrow \mathcal{B}(\beta_0, r_{\perp})$.

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We now apply these bounds to **indecision-breaking bifurcations on regular graphs**, which canonically model *consensus formation in opinion dynamics*.

NLOD Consensus Bifurcations: system

Consider the unbiased case for a k -regular graph:

$$\textbf{Hopfield} : \dot{x} = -dx + uA \tanh(x) \quad (2a)$$

$$\textbf{Firing Rate} : \dot{x} = -dx + \tanh(uAx) \quad (2b)$$

In both models, origin eqbm undergoes pitchfork bifurcation at $u^* = \frac{d}{k}$.

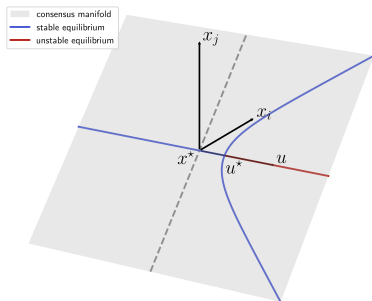


Figure: indecision-breaking bifurcation in (2) at $(x^*, u^*) = (\mathbf{0}, \frac{d}{k})$.

NLOD Consensus Bifurcations: bounds

LSR Validity Bounds for NLOD Consensus Bifurcation

Bounds of LSR validity for the above bifurcation are given by

- $0 < r_{\parallel} < \frac{d(k - \lambda_2(A))}{k|\lambda'(A)|}$ **along** consensus manifold
- $r_{\perp} > 0$ **normal** to consensus manifold

where $k = \lambda_1(A) \geq \lambda_2(A) \geq \dots$ and $|\lambda'(A)| := \max_{i \neq 1} \{|\lambda_i(A)|\}$.

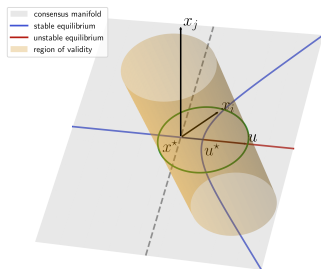


Figure: region of validity of LSR for (2).

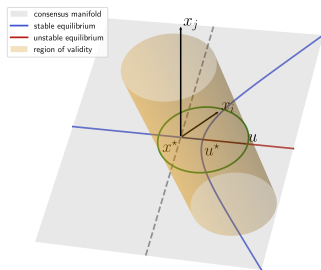
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Complete Graph: bounds simplify to $0 < r_{\parallel} < \frac{dn}{n-1}$ and $r_{\perp} > 0$ for the all-to-all complete n -graph case.

Figure: region of validity of LSR for (2).

NLOD Consensus Bifurcations (plots)

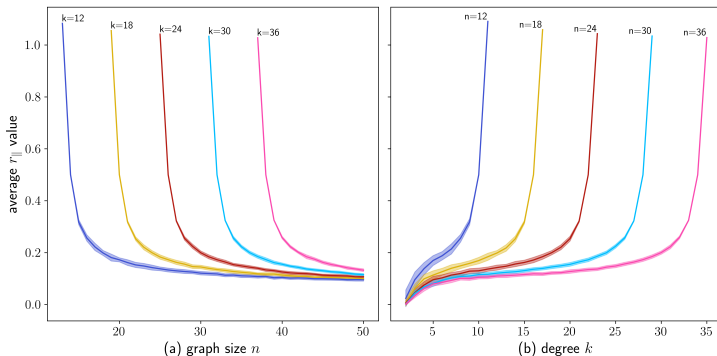


Figure: variation in bounds for NLOD over regular graphs ($d=1$):
(a) fixed degree $k \in \{12, 18, 24, 30, 36\}$, varying graph size n .
(b) fixed graph size $n \in \{12, 18, 24, 30, 36\}$, varying degree k .

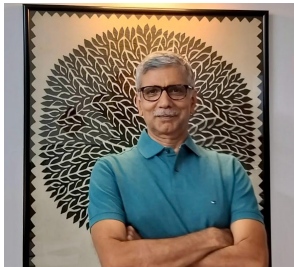
Summary of Contributions

- ① Developed computable first-order bounds for the region where LSR analysis remains faithful to the full system by leveraging explicit bounds on ImFT to obtain explicit neighbourhood radii $(r_{\parallel}, r_{\perp})$.
- ② Specialized general bounds to two broad nonlinear network models:
 - Hopfield-type dynamics: $\dot{x} = -Cx + pAS(x) + b$
 - Firing rate-type dynamics: $\dot{x} = -Cx + S(pAx + b)$
- ③ Applied bounds to indecision-breaking pitchfork bifurcations on consensus dynamics over k -regular graphs, revealing dependence on:
 - graph size n ,
 - network degree k ,
 - spectral structure of A .
- ④ Showed that Hopfield and Firing Rate NLOD yield *identical* validity bounds over regular graphs due to shared local Jacobian structure.

Thank You!



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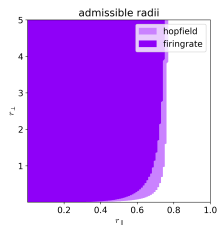
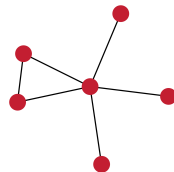
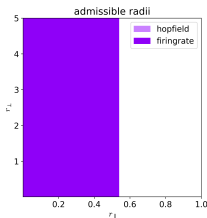
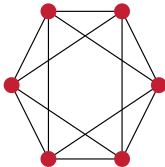
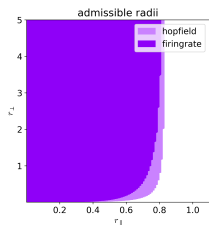
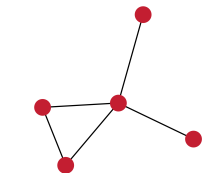
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- [GS85] M. Golubitsky and D. G. Schaeffer. *Singularities and Groups in Bifurcation Theory*. Vol. 51. Applied Mathematical Sciences. New York, NY: Springer-Verlag, 1985.
- [JCB24] Ashutosh Jindal, Debasish Chatterjee, and Ravi Banavar.
“Estimates of the size of the domain of the implicit function theorem: a mapping degree-based approach”. In: *Mathematics of Control, Signals, and Systems* 36.1 (2024), pp. 139–175.

Numerical Bounds for Non-regular Graphs



For arbitrary non-regular graph, bounds $r_{\parallel}$ and $r_{\perp}$ are coupled, and the bounds for the Hopfield and Firing Rate cases need not be identical.